

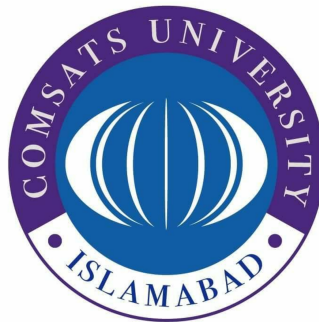
Assignment04: Public Key Infrastructure, Public Key Cryptography, Digital Signature Algorithm & One-Time Signatures

Submission deadline: 3-Dec-2025 08:29AM

Information Security

BS Computer Science

BCS4B



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1 System Architecture

1.1 Overall System Flow

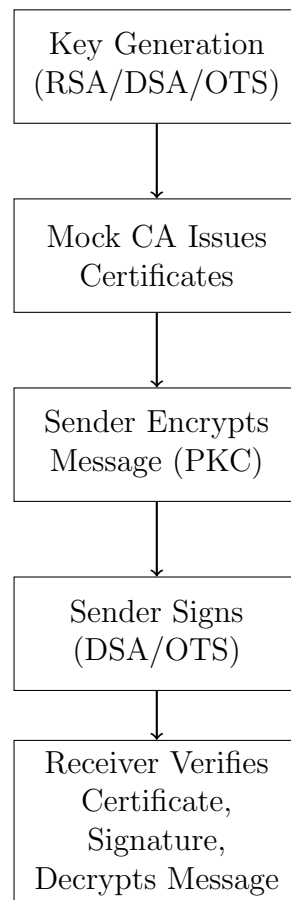


Figure 1: Secure Communication Pipeline

1.2 Communication Protocol

1. Setup Phase

- Alice and Bob generate RSA keys
- CA issues certificates for both
- Certificates exchanged publicly

2. Message Transmission

- Alice composes message M
- Alice computes: $C = E_{Bob}(M)$ (RSA encryption)
- Alice computes: $\sigma = \text{Sign}_{Alice}(M)$ (DSA/OTS)
- Alice sends $(C, \sigma, \text{Cert}_{Alice})$ to Bob

3. Reception & Verification

- Bob verifies Alice's certificate with CA

- Bob verifies signature σ using Alice's public key
- Bob decrypts: $M = D_{Bob}(C)$
- Communication assured to be: **Private, Authenticated, Integral**

2 Assignment Tasks

Task 1: RSA Implementation

Implement RSA encryption and decryption with the following:

1. Generate 1024-bit RSA key pair
2. Encrypt a plaintext message
3. Decrypt and verify recovery of original message
4. Test with at least 3 different messages

Task 2: PKI Certificate Management

Build a Mock Certificate Authority that:

1. Issues X.509-style certificates (simplified format)
2. Stores certificate metadata (issuer, subject, validity period)
3. Verifies certificate signatures using SHA-256
4. Implements certificate revocation (CRL)
5. Validates trust chains (optional: intermediate CA)

Task 3: Digital Signature Algorithm

Implement DSA signature and verification:

1. Generate DSA keys
2. Sign a message using private key
3. Verify signature using public key
4. Test with valid and invalid messages
5. Demonstrate non-repudiation property

Task 4: One-Time Signature Scheme

Implement Lamport OTS:

1. Generate OTS key pair (private: 256 pairs, public: 256 hashes)
2. Sign a 256-bit message hash
3. Verify OTS signature
4. Demonstrate that the key cannot be reused
5. Compare OTS security with DSA

Task 5: Integrated Secure Communication System

Build a complete system combining all four components:

1. Alice and Bob each have RSA, DSA, and OTS keys
2. CA issues certificates for both parties
3. Alice encrypts a message with Bob's public key
4. Alice signs the message with her private key (DSA + OTS)
5. Bob receives, verifies the certificate, verifies the signatures, and decrypts
6. Demonstrate three scenarios:
 - Successful communication
 - Message tampering detection
 - Certificate revocation